Double Integrals

ANSWER KEY

Double integrals over a rectangle

1 Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).
$\int_0^2 \int_0^3 xy\, dy\, dx$.

2 Set up the double integral of $f(x, y) = x^2 + y$ over the rectangle $[1, 3] \times [0, 2]$ (integrate x first).
$\int_0^2 \int_1^3 (x^2 + y)\, dx\, dy$.

Iterated integrals

3 Evaluate $\int_0^2 \int_0^3 xy\, dy\, dx$.
Inner: $\int_0^3 xy\, dy = x \cdot \frac{y^2}{2}\big|_0^3 = \frac{9x}{2}$. Outer: $\int_0^2 \frac{9x}{2}\, dx = \frac{9}{4}x^2\big|_0^2 = 9$.

4 Evaluate $\int_0^1 \int_0^2 (x^2 + y)\, dy\, dx$.
Inner: $\int_0^2 (x^2 + y)\, dy = x^2 y + \frac{y^2}{2}\big|_0^2 = 2x^2 + 2$. Outer: $\int_0^1 (2x^2 + 2)\, dx = \frac{2}{3}x^3 + 2x\big|_0^1 = \frac{8}{3}$.

5 Evaluate $\int_0^\pi \int_0^1 y \sin x\, dy\, dx$.
Inner: $\int_0^1 y \sin x\, dy = \sin x \cdot \frac{y^2}{2}\big|_0^1 = \frac{\sin x}{2}$. Outer: $\int_0^\pi \frac{\sin x}{2}\, dx = \frac{1}{2}[-\cos x]_0^\pi = \frac{1}{2}(1 - (-1)) = 1$.

Fubini's Theorem: order of integration

6 Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.
Order $dy\, dx$: inner $= x \cdot \frac{y^2}{2}\big|_0^2 = 2x$, outer $= \int_0^1 2x\, dx = 1$. Order $dx\, dy$: inner $= y \cdot \frac{x^2}{2}\big|_0^1 = \frac{y}{2}$, outer $= \int_0^2 \frac{y}{2}\, dy = 1$. Both orders give 1.

7 State the hypothesis of Fubini's Theorem needed for the two iterated orders to be guaranteed equal.
$f(x, y)$ must be continuous (or at least integrable) on the rectangular region $[a, b] \times [c, d]$ — under that condition, the double integral equals either iterated integral, in either order.

Double integrals over general regions

8 Evaluate $\int_0^1 \int_0^x (x + y)\, dy\, dx$.
Inner: $\int_0^x (x + y)\, dy = xy + \frac{y^2}{2}\big|_0^x = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2}\, dx = \frac{1}{2}x^3\big|_0^1 = \frac{1}{2}$.

9 Set up (do not evaluate) the double integral of $f(x, y) = xy$ over the triangular region bounded by $y = 0$, $x = 1$, and $y = x$.
$\int_0^1 \int_0^x xy\, dy\, dx$ (for a fixed x between 0 and 1, y ranges from the bottom edge $y = 0$ up to the line $y = x$).

10 Evaluate $\int_0^1 \int_y^1 2\,dx\,dy$ (a region bounded by $x = y$, $x = 1$, $y = 0$).

Inner: $\int_y^1 2\,dx = 2(1-y)$. Outer: $\int_0^1 2(1-y)\,dy = 2\left[y - \frac{y^2}{2}\right]_0^1 = 2\left(1 - \frac{1}{2}\right) = 1$.

Reversing the order of integration

11 Reverse the order of integration for $\int_0^1 \int_0^x f(x,y)\,dy\,dx$, describing the new bounds.

The region is $0 \le y \le x \le 1$. Reversed: for a fixed y between 0 and 1, x ranges from y up to 1: $\int_0^1 \int_y^1 f(x,y)\,dx\,dy$.

12 Evaluate $\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$ by first reversing the order of integration (the inner integral has no elementary antiderivative in the original order).

The region is $0 \le x \le y \le 1$. Reversed: $\int_0^1 \int_0^y e^{y^2}\,dx\,dy = \int_0^1 y\,e^{y^2}\,dy$. Substituting $u = y^2$, $du = 2y\,dy$: $= \frac{1}{2}\int_0^1 e^u\,du = \frac{1}{2}(e-1)$.

Polar coordinates

13 Convert $\iint_D (x^2+y^2)\,dA$ to polar coordinates, where D is the disk $x^2+y^2 \le 4$, and evaluate.

$x^2+y^2 = r^2$, $dA = r\,dr\,d\theta$. $\int_0^{2\pi}\int_0^2 r^3\,dr\,d\theta$. Inner: $\frac{r^4}{4}\big|_0^2 = 4$. Outer: $\int_0^{2\pi} 4\,d\theta = 8\pi$.

14 Evaluate $\iint_D xy\,dA$ where D is the quarter disk $x^2+y^2 \le 1$, $x \ge 0$, $y \ge 0$, using polar coordinates.

$xy = r^2 \sin\theta\cos\theta$, $dA = r\,dr\,d\theta$, $\theta \in [0, \frac{\pi}{2}]$, $r \in [0,1]$. $\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta \cdot \int_0^1 r^3\,dr = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}$.

15 Use polar coordinates to evaluate $\iint_D e^{-(x^2+y^2)}\,dA$ where D is the entire plane $\mathbb{R}^2$ (the classic Gaussian integral setup).

$\int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta$. Inner: let $u = r^2$, $du = 2r\,dr$: $\int_0^{\infty} e^{-r^2} r\,dr = \frac{1}{2}\int_0^{\infty} e^{-u}\,du = \frac{1}{2}$. Outer: $\int_0^{2\pi} \frac{1}{2}\,d\theta = \pi$.

Area and average value

16 Use a double integral to find the area of the region bounded by $y = x^2$ and $y = x$ (for $0 \le x \le 1$, where $x \ge x^2$).

Area $= \int_0^1 \int_{x^2}^x 1\,dy\,dx = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

17 The average value of $f(x,y)$ over a region D with area A is $\dfrac{1}{A}\iint_D f\,dA$. Find the average value of $f(x,y) = xy$ over the rectangle $[0,2]\times[0,3]$, using the earlier result $\iint_D xy\,dA = 9$.

Area of $[0,2]\times[0,3]$ is 6. Average value $= \dfrac{9}{6} = \dfrac{3}{2}$.

18 Use a double integral to find the area of the region bounded above by $y = 4 - x^2$ and below by $y = 0$.

The parabola meets $y = 0$ at $x = \pm 2$. Area $= \int_{-2}^2 \int_0^{4-x^2} 1\,dy\,dx = \int_{-2}^2 (4-x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^2 = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = 16 - \frac{16}{3} = \frac{32}{3}$.

Volume via double integrals

19 Set up (do not evaluate) the double integral for the volume under the surface $z = x^2 + y^2$ above the rectangle $[0, 1] \times [0, 1]$.
$V = \int_0^1 \int_0^1 (x^2 + y^2)\, dy\, dx$.

20 Evaluate the volume integral set up above.
Inner: $\int_0^1 (x^2 + y^2)\, dy = x^2 y + \frac{y^3}{3}\Big|_0^1 = x^2 + \frac{1}{3}$. Outer: $\int_0^1 \left(x^2 + \frac{1}{3}\right) dx = \frac{x^3}{3} + \frac{x}{3}\Big|_0^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

21 Find the volume of the solid bounded above by $z = 6 - x - y$ and below by $z = 0$, over the triangle in the xy-plane with vertices $(0, 0)$, $(6, 0)$, $(0, 6)$ (i.e., $0 \le x \le 6$, $0 \le y \le 6 - x$).
$V = \int_0^6 \int_0^{6-x} (6 - x - y)\, dy\, dx$. Inner: with $u = 6 - x$, $\int_0^u (u - y)\, dy = uy - \frac{y^2}{2}\Big|_0^u = u^2 - \frac{u^2}{2} = \frac{u^2}{2}$. Outer: $\int_0^6 \frac{(6-x)^2}{2}\, dx$. Substituting $u = 6 - x$, $du = -dx$: $= \int_0^6 \frac{u^2}{2}\, du = \frac{u^3}{6}\Big|_0^6 = \frac{216}{6} = 36$.

Double integrals in applications: mass and center of mass

22 A flat plate occupies the rectangle $[0, 2] \times [0, 1]$ with density $\rho(x, y) = x + y$. Set up the double integral for its total mass $M = \iint_D \rho\, dA$.
$M = \int_0^2 \int_0^1 (x + y)\, dy\, dx$.

23 Evaluate the mass integral set up above.
Inner: $\int_0^1 (x + y)\, dy = xy + \frac{y^2}{2}\Big|_0^1 = x + \frac{1}{2}$. Outer: $\int_0^2 \left(x + \frac{1}{2}\right) dx = \frac{x^2}{2} + \frac{x}{2}\Big|_0^2 = 2 + 1 = 3$. So $M = 3$.

24 The x-coordinate of the center of mass is $\bar{x} = \frac{1}{M}\iint_D x\rho\, dA$. Set up (do not evaluate) this integral for the plate above ($\rho(x, y) = x + y$ on $[0, 2] \times [0, 1]$, $M = 3$).
$\bar{x} = \frac{1}{3}\int_0^2 \int_0^1 x(x + y)\, dy\, dx$.

25 Evaluate $\bar{x}$ from the integral set up above.
Inner: $\int_0^1 x(x + y)\, dy = \int_0^1 (x^2 + xy)\, dy = x^2 y + \frac{xy^2}{2}\Big|_0^1 = x^2 + \frac{x}{2}$. Outer: $\int_0^2 \left(x^2 + \frac{x}{2}\right) dx = \frac{x^3}{3} + \frac{x^2}{4}\Big|_0^2 = \frac{8}{3} + 1 = \frac{11}{3}$. So $\bar{x} = \frac{1}{3} \cdot \frac{11}{3} = \frac{11}{9}$.